Ahmed Farag Ahmed

Machine Learning Engineer

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GitHub **LinkedIn** **portfolio**

PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer skilled in Python, SQL, data analysis, predictive modeling, and recommendation systems. Experienced in feature engineering, model evaluation, and ML deployment using FastAPI, with a focus on Deep Learning, NLP, Transformers, and LLMs.

EDUCATION

Alexandria University

Bachelor of Computer and Data Science | GPA: 3.43/4.00

Sep 2022 – Jul 2026

TECHNICAL SKILLS

Programming: Python, SQL, Java, C

Data Science: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing, Statistical Analysis, Data Visualization, Feature Engineering, Feature Selection, Data Transformation, Microsoft Excel, Power BI

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Ensemble Learning, Model Selection, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Deep Learning & AI: Neural Networks, Deep Learning, CNNs, Transfer Learning, TensorFlow, PyTorch, Keras

NLP: Text Preprocessing, TF-IDF, Text Classification, Word Embeddings, Transformers, LLM Fundamentals, RAG

Databases: MySQL, MongoDB, PostgreSQL, SQLAlchemy

Backend: FastAPI, REST APIs, JWT Authentication, Redis, Docker

Big Data: Apache Hadoop, HDFS, Apache Kafka, Apache Spark

Tools: Power BI, Microsoft Excel,Git, GitHub, VS Code, Jupyter Notebook

PROJECTS

1. Property Hub AI (Graduation Project)

AI-Powered Real Estate Platform — PropertyHUB

- Developed a full-stack AI-powered real estate platform using React, FastAPI, MySQL, and Redis.
- Built a hybrid recommendation system combining Content-Based Filtering and Learning-to-Rank techniques, trained using user interaction data from 200+ users.
- Developed a property price prediction model using 26,000+ real estate listings, achieving an R^2 score of 0.86.
- Designed and integrated RESTful APIs for authentication, property search, personalized recommendations, analytics, and price prediction.

2. Customer Churn Prediction

Machine Learning Prediction System

- Developed a machine learning solution to predict customer churn using Python and Scikit-learn on the Telco Customer Churn dataset.
- Performed data preprocessing, feature engineering, and class balancing using SMOTE on 7,043 customer records.
- Trained and evaluated five classification models, with Random Forest achieving the best performance at 80.3% accuracy.
- Evaluated model performance using Accuracy, Precision, Recall, and F1-score to support customer retention analysis.

PROFESSIONAL DEVELOPMENT

- Machine Learning Training — National Telecommunication Institute (NTI)
- AI & Machine Learning Summer Training — MAIM Company
- Computer Networks Training — Abu Qir Electricity Distribution Company
- IoT Summer Training — Faculty of Computers and Data Science